AMIT PAUL

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GitHub LeetCode LinkedIn Bitbucket Codeforces

Know more about me → https://amitpaul-portfolio.netlify.app/

Education

Khulna University | *B.Sc. Engg. in Electronics and Communication* Dec 2025

- **CGPA:** 3.18/4.00
- **Thesis:** A Comparative Study of Vision and Time Series Transformers for Bearing Fault Diagnosis Using PU Dataset.
- **Key Finding:** Analyzed Paderborn University real-world data, establishing that Vision Transformers provide maximum reliability (**99.28% accuracy**), while Time-Series models offer a computationally efficient alternative (**96.30%**).

Publications

Vision Transformer-Based Classification of Bearing Faults from Vibration Signal Spectrograms | *28th ICCIT, 2025*

- Proposed a methodology using Vision Transformers (ViT) on 2D spectrograms for bearing fault diagnosis, achieving 99.28% accuracy and outperforming standard CNN/RNN models in industrial predictive maintenance tasks.

Professional Experience

Idlewild Digital | Backend Developer (Remote) Jan 2024 – Present

- Architected "Crabarian" (export/import solution) and "Scholarstone" (LMS), automating workflows for 5,000+ users.
- Optimized database performance by resolving N+1 query issues and implementing efficient data retrieval strategies.

Idlewild Digital | Python Intern Aug 2023 – Dec 2023

- Owned the full lifecycle of the "Daily Updates" internal reporting system; promoted to Backend Developer due to rapid adoption of modular coding practices.

Internships

Grameenphone | Data Science & Engineering Trainee Jan 2025 – Feb 2025

- Designed a RAG-based chatbot which uses REST API to query telecom FAQs and serve real-time customer data.

Academic Projects

Dine Detectives (Smart Cafeteria System) | *Computer Vision, Arduino, Python*

- Developed an automated system using **Computer Vision** to track food consumption and integrated **Arduino-controlled gates** for automated billing.

University Certificate Automation System | *Django, Vue.js, PostgreSQL*

- Designed a full-stack certificate platform scalable to 10,000+ students; pitched architecture to University ICT Cell.

Digital Doctor Serial Management | *IoT, ESP32, MicroPython*

- Built a hybrid appointment system using ESP32 hardware handshake to synchronize SMS requests with a web portal.

Technical Skills

Deep Learning & AI: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
Computer Vision: OpenCV, Image Segmentation, PIL
IoT & Embedded: Arduino, ESP32, MicroPython, C++, Sensors & Actuators
Languages & Tools: Python, MATLAB, LaTeX, Git, Docker, Linux